A REPRESENTABLE MORPHISM WITH PROPERTY-P FIBRES OVER A LIMIT-PRESERVING BASE HAS PROPERTY P

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Editorial note. No omission markers in-span; proof complete as written.

Allowed standard prerequisites (editorial).

- Affines are limits of finite presentation. algebra-lemma-ring-colimit-fp (cross-chapter, named). Writes V = lim V_i after the fppf-local reduction.

Source reference. Original reference 00QN: focused-seeds/geometry/S9-crit-check-property-limit-preserving/ref-tags.txt, source lines 5–5. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 03YK: focused-seeds/geometry/S9-crit-check-property-limit-preserving/ref-tags.txt, source lines 6–6. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 0402: focused-seeds/geometry/S9-crit-check-property-limit-preserving/ref-tags.txt, source lines 7–7. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Lemma 0.1. *Let S be a scheme. Let $f : \mathcal{X} \to \mathcal{Y}$ be a 1-morphism of categories fibred in groupoids over $(Sch/S)_{fppf}$. Let $\mathcal{P}$ be a property of morphisms of algebraic spaces as in Algebraic Stacks, Definition 03YK. If*

(1) *f is representable by algebraic spaces,*
(2) *$\mathcal{Y} \to (Sch/S)_{fppf}$ is limit preserving on objects,*
(3) *for an affine scheme V locally of finite presentation over S and $y \in \mathcal{Y}_V$ the resulting morphism of algebraic spaces $f_y : F_y \to V$, see Algebraic Stacks, Equation (0402), has property $\mathcal{P}$.*

Then f has property $\mathcal{P}$.

Proof. Let V be a scheme over S and $y \in \mathcal{Y}_V$. We have to show that $F_y \to V$ has property $\mathcal{P}$. Since $\mathcal{P}$ is fppf local on the base we may assume that V is an affine scheme which maps into an affine open $\mathrm{Spec}(\Lambda) \subset S$. Thus we can write $V = \lim V_i$ with each V_i affine and of finite presentation over $\mathrm{Spec}(\Lambda)$, see Algebra, Lemma 00QN. Then y comes from an object y_i over V_i for some i by assumption (2). By assumption (3) the morphism $F_{y_i} \to V_i$ has property $\mathcal{P}$. As $\mathcal{P}$ is stable

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under arbitrary base change and since $F_y = F_{y_i} \times_{V_i} V$ we conclude that $F_y \to V$ has property $\mathcal{P}$ as desired. $\square$